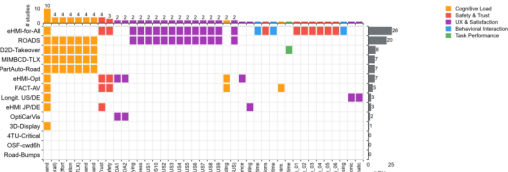


OpenDV-HCI: standardizing dependent variables turns fragmented open datasets into comparable, synthesizable evidence

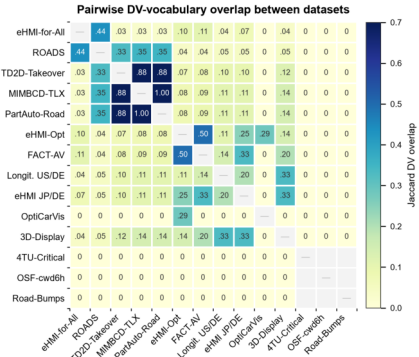
(a) DV coverage across 14 datasets

Dependent-variable coverage across 14 open HCI datasets



After schema-driven standardization, mean pairwise DV overlap is only Jaccard = 0.105. DVs appearing 0-3 studies: Mental Demand (n=10), NASA-TLX (overall) (n=4), Effort (n=4), Frustration (n=4), Performance (TLX) (n=4), >3 more: 3 sensoring datasets expose no canonical questionnaire DV.

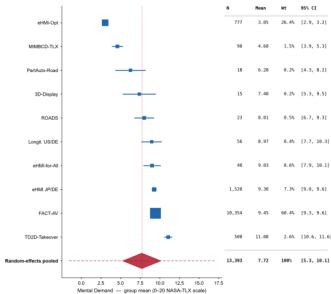
(b) Pairwise overlap (Jaccard)



(c) Meta-analysis: Mental Demand

Evidence synthesis unlocked by standardization: NASA-TLX Mental Demand

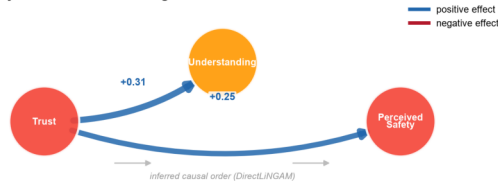
Pooling is only possible between heterogeneous source labels were mapped to same canonical DV. In a participant-level random-effects meta-analysis, high I^2 shows shared sources can still mask subpopulation differences.



Random effects (DerSimonian-Laird), $k = 10$ studies: heterogeneity: $I^2 = 96.8\%$, $\tau^2 = 14.47$, $Q(9) = 630.6$, $p < 0.001$

(d) Cross-study causal structure

Cross-study causal structure among standardized DVs



DirectLINGAM on pooled within-study z-scores (complete-case, non-synthetic; $n = 10,354$ obs). Edge weights are structural coefficients.